

Protected Attention–Koopman Takagi–Sugeno Fuzzy Models for Fault Detection and Margin-Certified Isolation in Closed-Loop Nonlinear Systems

Anonymous Authors

Abstract—Fault detection and isolation for closed-loop nonlinear processes often has to be designed from normal data alone. This paper develops a protected monitor around a learned attention–Koopman Takagi–Sugeno fuzzy model. Conditional on the recorded command path, its free-running reference has no direct post-anchor measurement-correction edge; closed-loop measurements can still affect it causally through those commands, and anchor cleanliness remains a declared coverage event. The stacked residual admits a first-order decomposition into anchor propagation, path-dependent mismatch, stochastic disturbance, and conditional fault signatures. Three coupled constructions follow. First, a white-space spectral quotient partially removes the anchor nuisance, prices its retained component with the same deployed operator used by each statistic, and adds an unquotiented guard. Second, an input- and path-scheduled threshold propagates compact nuisance sets and calibrates a frozen superblock-maximum score, providing a simultaneous guarantee for a prespecified finite episode rather than an indefinite pointwise claim. Third, a set-valued isolation layer compares raw-record-consistent actuator, sensor, process, mixed, and nuisance-only explanation sets; channel masks enter as counterfactual recomputations, not new physical observations. Detection and attribution use nested normal regions, and a fault label is issued only when the larger attribution-normal region and every competing fault explanation are excluded on separate signature- and nuisance-coverage events. Coincident supports rule out support-wise robust labels; quantified class risk additionally requires explicit signature and nuisance budgets and is not posterior confidence.

Index Terms—Fault detection, fault isolation, Takagi–Sugeno fuzzy systems, Koopman operator, dynamic threshold, conformal calibration, closed-loop systems.

I. INTRODUCTION

SAFETY-critical industrial processes operate in closed loop, traverse multiple nonlinear operating regimes, and must be monitored for actuator, sensor, and process faults whose signatures are not known in advance. In most applications the data available for monitor design are records of normal operation only: faults are rare, expensive to reproduce, and almost never labeled. A practical diagnosis scheme therefore has to learn what normal closed-loop behavior looks like, detect departures from it with a controlled false-alarm rate, and attribute a detected departure to a structural fault class—all without ever having seen a fault.

Three methodological traditions address parts of this problem. Model-based residual generation—observer-based

and parity-space design, residual evaluation, and threshold computation—provides the most complete theory [1]–[5], but presupposes an explicit dynamic model with a priori uncertainty bounds. Multivariate statistical monitoring builds data-driven normal-behavior models, yet its classical instruments are static projections that ignore closed-loop dynamics [6], [7]. Between the two, learned dynamic models have emerged: Koopman-operator liftings represent nonlinear dynamics by (bi)linear evolution in an observable space [8]–[11], T–S fuzzy structures preserve local-linear models under nonnegative memberships that sum to one [12], and attention encoders summarize input–output histories into latent states [13]. Recent work applies learned Koopman models to fault diagnosis with parity or observer residuals trained on fault-free data [14]–[16].

The first unresolved difficulty is reference contamination. Every online monitor corrects its normal-behavior reference with incoming measurements—through an observer gain, a parity stack over the latest window, or an encoder that re-reads the measurement history. After a fault occurs, the same channel can drag the reference toward the faulty plant. The opposite extreme—a permanently free-running simulation—removes that direct correction edge but accumulates prediction error [17]. In closed loop it is not causally independent of measurements, because the controller-generated recorded commands may carry their effects. What is needed is an architecture that separates the two needs in time: delayed no-alarm candidate anchors, no direct post-anchor measurement correction conditional on the recorded commands, and an explicit clean-anchor event rather than an unobservable claim that delayed acceptance proves fault freedom.

The second difficulty concerns the evaluation of stacked multi-step residuals. When the reference is a free-running rollout, the natural residual is the whole window of multi-step prediction errors, and its fault-free value is driven by propagated anchor uncertainty, input-dependent model mismatch, and serially correlated noise. Classical dynamic thresholds propagate a priori uncertainty bounds through known LTI or LPV residual dynamics [18]–[21]; set-based thresholds propagate zonotopes or intervals through given analytic T–S or LPV models [22]–[26]. None of these constructions applies when the model itself is learned, the mismatch envelope must be estimated from normal data as a function of the measured input, and the residual is a stacked

window whose rows share the same disturbance realizations. Conversely, conformal methods deliver distribution-free false-alarm control for arbitrary scores [27]–[30], but calibrate the raw score directly and discard the model structure that makes a threshold input-aware. A principled division of labor between a model-based deterministic threshold generator and a distribution-free probabilistic layer is missing.

The third difficulty is structured isolation. Structured and directional residuals, incidence matrices, and unknown-input observer banks solve isolation when fault entry maps are known [2], [31]–[33]; data-driven constructions exist for linear systems [14], [34]; learned-model diagnosis either isolates within a single fault category [15] or relies on empirical channel masking without guarantees [35]. Quantitative isolability has been studied through distinguishability measures and residual-direction limits [36], [37], and principal angles between actuator and sensor signature subspaces have recently been used for nonlinear systems with learned observer components [38], though without a process class, an only-normal-data protocol, or rejection semantics. What is absent is a class-level theory—actuator versus sensor versus process—for residuals of learned closed-loop models: explicit signature operators, conditions under which the classes are distinguishable, a margin that can be evaluated online, and an honest rejection output when the margin fails.

This paper addresses the three difficulties jointly, for closed-loop nonlinear systems and under a strict only-normal-data protocol. The key observation is that all three reduce to properties of one object: the stacked residual of a protected free-running rollout, together with a small family of structured operators—anchor-error propagation, innovation propagation, and fault signatures—that are represented online by Jacobian-vector products of the learned model and, whenever used for safe exclusion, by certified outer enclosures. The contributions are:

- 1) *Protected residual generation.* A dual-branch architecture in which a data branch encodes the strict past of the measured closed loop while a protected branch rolls the learned attention–Koopman T–S model freely from a delay-confirmed anchor. A state machine removes direct post-anchor measurement correction conditional on the recorded-command path; it does not remove the causal measurement–controller–command path. Clean-anchor status itself remains an explicit event because absence of a prior alarm does not exclude a pre-anchor persistent fault.
- 2) *Post-filter design for the stacked residual.* A first-order structural decomposition of the stacked fault-free residual is derived, with an explicitly quantified bilinear remainder. On its basis a three-stage post-filter is designed: a dimensionless, energy-selective quotient of the anchor-error subspace, cross-time whitening, and a branch bank with an unannihilated guard and low-dimensional matched statistics [39]. The retained anchor component is priced in each deployed, whitened branch, and the information loss is expressed by an exact retained-signature fraction rather than an unsupported

onset monotonicity claim.

- 3) *Input- and path-scheduled dynamic threshold.* An online threshold generator propagates normal-data-estimated mismatch sets through the realized Jacobian path and adds the branchwise residual-anchor radius. A held-out block calibration prices the observable excess only after the post-filter, modes, masks, and scores are frozen. The result separates deterministic set containment, finite-horizon selection-safe calibration, and fault-time detectability assumptions instead of transferring a normal false-alarm statement to the fault regime.
- 4) *Set-valued structured isolation.* Actuator, sensor, and process explanations are represented by operating-path-conditioned reachable sets. The full diagnostic output combines the linear geometry of simultaneously computed residual branches with the counterfactual product output of mask-conditioned recomputations. It returns all feasible classes or supports, and reports a unique label only when every competing explanation tube is separated; physical actuator and process labels are explicitly conditional on input excitation and externally supplied process-injection structure.

Throughout, the closed-loop information boundary is stated explicitly. Conditioning both branches on the same recorded command cancels the command itself, but it does not cancel an unrecorded additive actuator or process injection. Controller feedback changes detectability only through the realized operating path, saturation, scheduling, and model mismatch; for sensor faults, the recorded-command path cancels only under an exact command-conditioned model.

The remainder of the paper is organized as follows. Section II reviews controlled Koopman liftings and T–S fuzzy structure, states the system and information model, and formulates the two design objectives. Section III develops the protected architecture, the structural decomposition and post-filter, the dynamic threshold generator, the isolation layer, and the learning procedure. Section IV specifies the evaluation protocol on a closed-loop continuous stirred-tank reactor benchmark. Section V concludes. Long proofs are collected in the Appendix.

Notation: $\mathbb{R}^m$ denotes m -dimensional Euclidean space and $\|\cdot\|$ the Euclidean norm or the induced spectral norm. Dimensions carry the prefix m (e.g., m_z), k indexes samples, and s denotes an anchor instant. Boldface lowercase letters are stacked vectors and boldface capitals stacked operators, displayed with explicit brackets at first use. Calligraphic capitals denote mappings, and blackboard-bold capitals denote sets, spaces, and subspaces. The fault family is reserved for f and its capital forms; noise and disturbance terms use ε with roman superscripts. For a matrix M , M^T , $M^\dagger$, $\sigma_{\min}(M)$, and $\text{Range}(M)$ are its transpose, pseudoinverse, smallest singular value, and column space. I_m is the $m \times m$ identity, and Π_V the orthogonal projector onto a subspace V .

II. PRELIMINARIES AND PROBLEM FORMULATION

A. Controlled Koopman Lifting

Consider a discrete-time controlled nonlinear system $x_{k+1} = \mathcal{X}(x_k, u_k, \xi_k)$ with state $x_k \in \mathbb{R}^{m_x}$, control input

$u_k \in \mathbb{R}^{m_u}$, and measured exogenous condition $\xi_k \in \mathbb{R}^{m_\xi}$. Let $\mathbb{H}^\infty$ be a Hilbert space of observables $\psi : \mathbb{R}^{m_x} \rightarrow \mathbb{R}$. The controlled Koopman operator family $\mathcal{K}_{(u,\xi)} : \mathbb{H}^\infty \rightarrow \mathbb{H}^\infty$ acts by composition with the frozen-input transition,

$$(\mathcal{K}_{(u,\xi)}\psi)(x) = \psi(\mathcal{X}(x, u, \xi)), \quad (1)$$

so that the nonlinear state evolution becomes linear evolution of observables for each frozen input pair [8]. Finite-dimensional computation replaces (1) by a controlled linear predictor on a learned lifting $\Psi : \mathbb{R}^{m_x} \rightarrow \mathbb{R}^{m_z}$,

$$z_{k+1} = Az_k + B_u u_k + B_\xi \xi_k, \quad \hat{y}_k = C_z z_k, \quad (2)$$

with $z_k \approx \Psi(x_k) \in \mathbb{R}^{m_z}$, matrices $A \in \mathbb{R}^{m_z \times m_z}$, $B_u \in \mathbb{R}^{m_z \times m_u}$, $B_\xi \in \mathbb{R}^{m_z \times m_\xi}$, and an output map $C_z \in \mathbb{R}^{m_y \times m_z}$; (2) is a genuine approximation, obtained in practice by extended dynamic mode decomposition or by jointly learning Ψ and the matrices [9]–[11]. A single linear predictor, however, cannot represent strongly regime-dependent dynamics, which motivates the fuzzy combination reviewed next.

B. T-S Fuzzy Combination of Local Koopman Models

A T-S fuzzy model [12] interpolates N_{rule} local models through memberships defined on a premise variable $\rho \in \mathbb{R}^{m_\rho}$. Throughout we require the memberships $\omega_i(\rho)$ to satisfy the standard conditions

$$\omega_i(\rho) \geq 0, \quad \sum_{i=1}^{N_{\text{rule}}} \omega_i(\rho) = 1. \quad (3)$$

In this paper the memberships are *learned*: a premise network $\mathcal{P}$ maps the available latent information to ρ , rule scores are quadratic in ρ with learnable prototypes and bounded-spectrum metrics, and ω_i is their softmax, so (3) holds by construction and all maps are continuously differentiable. The T-S Koopman transition used throughout is

$$\mathcal{Z}_\Theta(z, u, \xi; \chi) = \sum_{i=1}^{N_{\text{rule}}} \omega_i(\rho) (A_i z + B_i u + o_i), \quad (4)$$

where $\rho = \mathcal{P}(z, \chi, u, \xi)$, the pair (A_i, B_i, o_i) is the i th local Koopman model, and $\chi \in \mathbb{R}^{m_\chi}$ is an attention context defined in Section III. The complete Jacobians of (4), including the membership-variation term, are given in Section III-B; they are the objects propagated by the threshold generator.

C. System, Faults, and Information Model

Assumption 1 (Plant and fault channels). The monitored closed-loop plant evolves as

$$\begin{aligned} u_k^{\text{act}} &= u_k + f_k^a, \\ x_{k+1} &= \mathcal{X}^n(x_k, u_k^{\text{act}}, \xi_k) + f_k^p + \varepsilon_k^x, \\ y_k &= Cx_k + f_k^s + \varepsilon_k^y, \end{aligned} \quad (5)$$

where $\mathcal{X}^n$ is the unknown fault-free transition mapping, u_k is the command recorded by the controller, $y_k \in \mathbb{R}^{m_y}$ the measured output, $f_k^a \in \mathbb{R}^{m_u}$, $f_k^p \in \mathbb{R}^{m_x}$, and $f_k^s \in \mathbb{R}^{m_y}$ are actuator, process, and sensor faults, and $\varepsilon_k^x, \varepsilon_k^y$ are process

disturbance and measurement noise. In fault-free operation $f_k^a = f_k^p = 0$ and $f_k^s = 0$.

Assumption 2 (Information boundary). Online, the monitor trusts the recorded command u_k , the measurement y_k , and the exogenous condition ξ_k ; the applied input u_k^{act} , the state x_k , and all fault signals are unavailable. The exogenous condition is assumed unaffected by faults. Model training, structure and hyperparameter selection, and all threshold calibration use normal data only, partitioned into disjoint training, estimation, detection-calibration, attribution-calibration, and frozen-test blocks; fault data enter only the final frozen test.

Assumption 3 (Normal operating domain). The learned model, once frozen, is continuously differentiable on a compact normal latent domain, and the fault-free forcing terms defined in Section III-B admit envelopes estimated on normal validation data. All guarantees are stated conditionally on these envelopes and are flagged as such.

The closed-loop setting deserves emphasis. The controller maps measurements to commands, so a sensor fault also perturbs future commands. Because the monitor conditions *both* branches on the same recorded command sequence, that recorded command is common to the two model evaluations. This does not remove the unrecorded terms f_k^a or f_k^p in (5): in an exact additive linearization the residual recursion contains $B_k^u f_k^a + D_k^p f_k^p$ even when the controller changes u_k . The shared-command path of a sensor fault cancels only for an exact command-conditioned model; otherwise its leading leakage is generally first order in the command-dependent model error. Feedback can still alter all signatures through the realized operating point, saturation, scheduling, and unmodeled actuator dynamics.

D. Problem Formulation

Let $\mathcal{D}_n = \{(u_k, y_k, \xi_k)\}$ denote the normal dataset of Assumption 2, and let $\alpha \in (0, 1)$ be a prescribed false-alarm level. The paper solves the following two problems.

Problem 1 (*protected finite-episode detection*). Design, using $\mathcal{D}_n$ only, a residual generator with no direct post-anchor measurement-correction edge conditional on the recorded command path, a finite branch bank, and path-scheduled thresholds. After freezing the monitor state, candidate, modes, and score map, require the maximum over every selectable statistic in a prespecified fault-free episode to satisfy

$$\Pr \left\{ \max_{k \text{ in episode, } b} \frac{(T_{b,k} - \gamma_{b,k}^{\text{anc}} - \gamma_{b,k}^{\text{det}})_+}{\sigma_b} > q_{1-\alpha}^{\text{det}} \right\} \leq \alpha, \quad (6)$$

and give fault-time detectability only conditional on an independently stated signature and nuisance event.

Problem 2 (*risk-limited set-valued isolation*). Construct full-monitor-state/raw-record-consistent explanation sets $\mathbb{E}_c$ for $c \in \{0, \text{a}, \text{s-1}, \dots, \text{s-}m_y, \text{p}, \text{mixed}\}$, where 0 is nuisance-only. Given the frozen sufficient deployed observation $\mathcal{O}_{\text{obs}}^{\text{full}}$, output

$$\mathcal{C}_k^+ = \{c : \mathcal{O}_{\text{obs}}^{\text{full}} \in \mathbb{E}_c^{\text{full}}\}. \quad (7)$$

A physical fault label is permitted only when $\mathcal{C}_k^+$ is a nonnormal singleton; otherwise the monitor reports NORMAL-COMPATIBLE, NONUNIQUE, an equivalence class, or OUT-OF-MODEL.

III. PROPOSED METHOD

A. Protected Dual-Branch Residual Generation

The monitor maintains two strictly separated information paths built on one learned model $\{\mathcal{E}, \mathcal{P}, \mathcal{Z}_\Theta, C_z\}$.

1) *Data branch*: At every instant the causal attention encoder $\mathcal{E}$ reads the strict-past history

$$d_k^- = [u_{k-\ell_h}^\top \ y_{k-\ell_h}^\top \ \cdots \ u_{k-1}^\top \ y_{k-1}^\top \ \xi_{k-\ell_h}^\top \ \cdots \ \xi_k^\top]^\top \quad (8)$$

and returns the data-branch latent state and attention context,

$$(z_k^E, \chi_k^E) = \mathcal{E}(d_k^-) \in \mathbb{R}^{m_z} \times \mathbb{R}^{m_x}. \quad (9)$$

The current measurement y_k is deliberately excluded from d_k^- : a sensor abnormality at instant k cannot enter the latent state before $k+1$, the current output residual retains direct sensitivity to it, and the isolation layer of Section III-D obtains a clean temporal ordering of fault propagation.

2) *Protected branch*: Let s be a *candidate anchor*, an instant accepted by the delayed no-alarm rule below. The protected branch initializes $\hat{z}_{s|s} = z_s^E$ and rolls the model freely,

$$\hat{z}_{t+1|s} = \mathcal{Z}_\Theta(\hat{z}_{t|s}, u_t, \xi_t; \chi_{t|s}^n), \quad \hat{y}_{t|s} = C_z \hat{z}_{t|s}, \quad (10)$$

where the protected context $\chi_{t|s}^n$ is produced by $\mathcal{E}$ from a protected history buffer that contains the *predicted* outputs $\hat{y}_{\cdot|s}$ together with the recorded u, ξ ; no measurement taken after s enters a measurement slot of (10). Thus, for $s < \ell < t$,

$$\left. \frac{\partial \hat{z}_{t|s}}{\partial y_\ell} \right|_{u_{s:t-1}, \xi_{s:t-1}} = 0, \quad (11)$$

whereas the total closed-loop derivative need not vanish because y_ℓ may change a later recorded command. This proves recorded-command-conditional direct noncontamination, not causal independence or anchor cleanliness. The recorded command is common to both model evaluations, while unrecorded additive actuator and process injections remain in the residual as stated in Section II.

3) *Anchor state machine*: A candidate anchor at instant s is accepted only at instant $s+N_c$, and only if a dedicated anchor-eligibility score remains below a threshold q_{anc} throughout $(s, s+N_c]$. The score map, q_{anc} , hysteresis, reset state, and re-anchoring schedule are selected on the training/estimation blocks and frozen *before* final conformal calibration. The final detection quantile produces an alarm output but never changes the anchor, mode, or window path that generates the scores being calibrated. Let N_a be a separately declared uniform delay within which this frozen eligibility gate becomes and remains unsafe over a specified fault family with minimum amplitude/growth conditions; it is not implied by network order.

Proposition 1 (Conditional direct-path blocking). *If the fault family satisfies the stated finite N_a anchor-gate assumption*

and $N_c \geq N_a$, then no candidate containing a post-onset measurement token is newly accepted after the gate's reaction. Equation (11) holds for every accepted rollout. The proposition neither blocks the causal path $y \rightarrow u \rightarrow \hat{z}$, nor certifies that an older anchor was clean; arbitrarily slow faults have no finite bound without a minimum growth condition.

The proof is immediate from the construction and is omitted. The choice of N_c is a design assumption tied to the declared fault family and frozen anchor gate; it cannot be derived from ℓ_h or m_z alone. Using the final calibration quantile inside that gate would make the score path and its order statistic circular and invalidate the split-conformal rank argument.

4) *Unified residual and exact propagation*: For any start instant s (a training start, the short-horizon start $k-\ell_s$, or the long-term anchor s_k^n), define the residual, the one-step innovation, and the context mismatch

$$\begin{aligned} e_{k|s} &= z_k^E - \hat{z}_{k|s}, \\ \nu_k &= z_{k+1}^E - \mathcal{Z}_\Theta(z_k^E, u_k, \xi_k; \chi_k^E), \\ \delta_{k|s}^x &= \mathcal{Z}_\Theta(z_k^E, u_k, \xi_k; \chi_k^E) - \mathcal{Z}_\Theta(z_k^E, u_k, \xi_k; \chi_{k|s}^n), \end{aligned} \quad (12)$$

together with the output residual $r_k^y = y_k - C_z z_k^E$. All four quantities are online computable, and the same rollout (10) is used in training and online, which removes the distribution mismatch between teacher-forced training and free-running monitoring.

Proposition 2 (Exact stacked propagation). *Let $\mathcal{P}$ be continuously differentiable, and define the segment-averaged Jacobian and transition products*

$$\begin{aligned} A_{k|s} &= \int_0^1 J_z \mathcal{Z}_\Theta(\hat{z}_{k|s} + q e_{k|s}, u_k, \xi_k; \chi_{k|s}^n) dq, \\ \Phi(k, j) &= A_{k-1|s} \cdots A_{j|s}, \quad \Phi(k, k) = I_{m_z}. \end{aligned} \quad (13)$$

Then, for every $l \geq 1$,

$$e_{s+l|s} = \sum_{j=0}^{l-1} \Phi(s+l, s+j+1) (\nu_{s+j} + \delta_{s+j|s}^x), \quad (14)$$

an exact identity that holds for arbitrary fault action on the data branch, because no fault quantity was differentiated in its derivation.

Stacking the window rows of (14) gives the central object of the paper. With $e_{s,N} = [e_{s+1|s}^\top \cdots e_{s+N|s}^\top]^\top \in \mathbb{R}^{Nm_z}$ and ν, δ stacked likewise,

$$e_{s,N} = G_\nu (\nu + \delta), \quad [G_\nu]_{l,j} = \begin{cases} \Phi(s+l, s+j), & j \leq l, \\ 0, & j > l, \end{cases} \quad (15)$$

where $[G_\nu]_{l,j} \in \mathbb{R}^{m_z \times m_z}$ denotes the (l, j) block. Since the diagonal blocks equal I_{m_z} , the operator G_ν is unit block lower triangular and invertible: the stacked residual and the innovation sequence carry the same information, and the choice between them is a choice of coordinates. The complete fuzzy Jacobians needed in (13), including the membership-variation term $\Delta_\rho J_{\rho,z}$ that dominates near rule boundaries, are assembled analytically from the softmax gradient identity

and are stated with their dispersion bounds in the Appendix (Proposition 6); their neglect systematically underestimates the propagation gain, which is why all operators in this paper are built from the complete Jacobians.

The integral in (13) is an exact analytical identity, not automatically an exactly evaluated online quantity. Deployment uses a certified enclosure

$$A_{k|s} \in \mathbb{A}_{k|s} = \{\hat{A}_{k|s} + \Delta : \|\Delta\| \leq \epsilon_{k|s}^A\}, \quad (16)$$

obtained by interval evaluation or quadrature with a verified remainder. Products, block-column norms, and signature maps are propagated as outer enclosures. The same operator-error element is shared across all branches and class explanations: its detection image is included in γ^{det} , and its class-dependent image is included in the synchronized explanation tube. Without a certified $\epsilon_{k|s}^A$, Proposition 2 remains a conceptual identity, but the deterministic containment and safe-exclusion claims are unavailable.

B. Structural Decomposition and Post-Filter Design

This subsection derives the structure of the fault-free stacked residual and designs the evaluation filter on it. Throughout, s is the window anchor and N the window length; nominal operators are assembled online along the protected trajectory from Jacobian–vector products of the frozen model, while deterministic certificates use the enclosures in (16).

1) Context cancellation and the combined forcing:

Lemma 1 (Context cancellation). *For every k and start s , $\nu_k + \delta_{k|s}^X = z_{k+1}^E - Z_\Theta(z_k^E, u_k, \xi_k; \chi_{k|s}^n)$: the dependence on the data-branch context χ_k^E cancels algebraically, at every fault amplitude.*

The proof is one line from (12). The stacked identity (15) therefore depends on the data branch only through the latent sequence z^E and on the protected branch through the normal context; in particular, fault energy appearing in δ^X alone is an artifact of the additive split and carries no isolation information (Section III-D).

2) *First-order structure of the fault-free window:* Write $z_k^E = z_k^* + \epsilon_k^e$, where z^* is the noise-free encoding of the underlying normal trajectory and ϵ_k^e collects the measurement- and encoder-induced per-step error, including the context-noise term $J_\chi Z_\Theta \cdot (\partial\chi/\partial Y) \epsilon^y$ of the data branch. Expanding the combined forcing of Lemma 1 to first order and telescoping the ϵ^e chain through G_ν yields the following.

Proposition 3 (Structural decomposition with quantified remainder). *Let L_H be a Lipschitz constant of $z \mapsto J_z Z_\Theta$ on the normal domain. In fault-free operation,*

$$e_{s,N} = \epsilon_+^e - G_0 \epsilon_s^e + G_\nu (\epsilon^m + \epsilon^v + d^X) + r^{\text{rem}}, \quad (17)$$

where $\epsilon_+^e = [(\epsilon_{s+1}^e)^\top \dots (\epsilon_{s+N}^e)^\top]^\top$ carries the per-step encoding errors, $G_0 = [\Phi(s+1, s)^\top \dots \Phi(s+N, s)^\top]^\top \in \mathbb{R}^{Nm_z \times m_z}$ propagates the anchor-instant error through the whole window, $\epsilon^m, \epsilon^v, d^X$ are the stacked model-mismatch, process-noise, and context-drift forcings, and the remainder

obeys $\|r^{\text{rem}}\| = O(L_H (\|\epsilon^e\|^2 + \|\epsilon^e\| \|\epsilon\|))$. Under active faults, the window gains the signature terms

$$e_{s,N} += F_a f^a + \sum_{j=1}^{m_y} F_s^{(j)} f_j^s + G_\nu w^p, \quad (18)$$

with $F_a = G_\nu \text{blkdiag}(B_s^u, \dots, B_{s+N-1}^u)$ the transition-weighted input-response operator, $F_s^{(j)}$ the banded operator assembled from the encoder Jacobian–vector products with respect to the channel- j measurement-history slots (including, for pre-anchor onsets, the protected-context-buffer slots), and w^p an unknown state-channel forcing representing process faults.

Three structural facts in (17)–(18) drive everything that follows. First, the anchor-error term $-G_0 \epsilon_s^e$ occupies a subspace of dimension at most m_z , is perfectly correlated across all window rows, and grows with the transition products—it is precisely the term whose scalar upper bound dominates radius-recursion thresholds at long reference ages. Second, actuator and process faults are *propagated* forcings, filtered by G_ν , whereas sensor faults enter through the encoder image of the history as *banded* perturbations of width ℓ_h that do not accumulate along the window; this asymmetry, a direct consequence of the protected branch never reading post-anchor measurements, is the structural basis of isolation. Third, the remainder is *bilinear*: its constant grows with the sharpness of the rule metrics (through L_H), so the decomposition—and every statement built on it—carries the explicit validity condition that the metric sharpness times the normal error envelope be small; the monitor evaluates a membership-dispersion gate online and degrades to the guard statistic below whenever the gate fails. In the numerical verification of Section IV, the identity (15) holds to 10^{-15} , the anchor term is first-order exact with the predicted bilinear scaling, and the banded sensor structure is reproduced with machine-zero out-of-band response.

3) Stage 1: anchor annihilation:

Proposition 4 (Anchor quotient and retained signature). *rank $G_0 = \text{rank } A_s$, where A_s is the pointwise Jacobian at the anchor (the segment average degenerates to it because $e_{s|s} = 0$). Let $Q \in \mathbb{R}^{Nm_z \times (Nm_z - r_0)}$, $r_0 = \text{rank } G_0$, be an orthonormal basis of $\text{Range}(G_0)^\perp$. Then $Q^\top e$ is free of the anchor-error term up to the bilinear remainder. A signature g is invisible to every fully annihilated statistic if and only if $g \in \text{Range}(G_0)$, and its exact retained Euclidean fraction is*

$$\rho(g) = \frac{\|Q^\top g\|^2}{\|g\|^2} = 1 - \frac{\|U_0^\top g\|^2}{\|g\|^2}, \quad (19)$$

where U_0 is an orthonormal basis of $\text{Range}(G_0)$. For a first-post-anchor actuator impulse, $\text{Range}(F_a^{(1)}) \subseteq \text{Range}(G_0)$ holds if and only if $\text{Range}(B_s^u) \subseteq \text{Range}(A_s)$; it is therefore automatic for nonsingular A_s , but not for a rank-deficient anchor Jacobian. No general onset monotonicity of ρ is claimed for time-varying propagation.

The annihilation is therefore a deliberate and quantified trade. For the deployed partial quotient, whitening and

selection must be defined in the same coordinates. Let $W_0 = \Sigma_e^{-1/2}$ and form

$$K_{\text{anc}} = W_0 \mathbf{G}_0 R_\eta^{1/2}. \quad (20)$$

Let $\mathbb{U}_1$ be the selected left singular subspace of K_{anc} , let Q_w span $\mathbb{U}_1^\perp$, and define the common retained operator $L_0 = Q_w^\top W_0$. If Σ_e includes the anchor variation itself, (20) measures anchor strength relative to total normal variation, not an anchor-to-noise ratio. A singular-value cut τ is implemented as the spectral projector $\mathbf{1}_{(\tau, \infty)}(K_{\text{anc}} K_{\text{anc}}^\top)$ and is used only with a gap; equivalently, an eigenvalue cut λ obeys $\sigma_\tau^2 > \lambda > \sigma_{\tau+1}^2$. A repeated singular-value cluster is retained or removed as a whole, and a threshold collision is treated as a separate hybrid mode. Partial annihilation leaves a nonzero anchor component. For branch b with $L_b = P_b L_0$, its deployed-coordinate radius

$$\gamma_{b,k}^{\text{anc}} = \|L_b \mathbf{G}_0 R_\eta^{1/2}\| \quad (21)$$

must be added to the threshold. Projecting the white-space singular vectors directly in the raw residual coordinates would not annihilate the selected direction. The bank also retains a guard with $L_g = I$ to observe directions discarded by the quotient. Thus full annihilation gives exact anchor-translation invariance, whereas partial annihilation gives only a subgroup invariant plus the priced residual (21).

4) *Stage 2: cross-time whitening*: The covariance Σ_e of the stacked fault-free residual is estimated once per fuzzy region on the estimation block with Ledoit–Wolf shrinkage, a ridge, and an eigenvalue floor [40]; the resulting W_0 is used in the same frozen candidate that defines K_{anc} , Q_w , and every L_b . Serial correlation of the encoding errors is absorbed by Σ_e itself. At the population level, any invertible prefilter followed by exact rewhitening differs from direct whitening only by an orthogonal transformation, and hence cannot create fault information or improve whitened class separation. Finite-sample differences may arise because regularization is not generally equivariant to arbitrary coordinate changes; these are reported as regularization effects, not information gains.

5) *Stage 3: branch bank*: Detection uses a bank of statistics on each window, all computed unconditionally at every step so that their calibration on normal data requires no alarm-conditioning:

$$\begin{aligned} T_{0,k} &= \|L_0 e\| && \text{(omnibus, annihilated),} \\ T_{g,k} &= \|e\| && \text{(guard, un-annihilated),} \\ T_{c,k} &= \|P_c L_0 e\| && \text{(matched, } c \in \mathcal{C}), \end{aligned} \quad (22)$$

Here $P_0 = I$ and each P_c is a frozen orthogonal projector in the retained white-space coordinates. The complete branch operators are $L_0 = Q_w^\top W_0$, $L_c = P_c L_0$, and $L_g = I$. The finite library $\mathcal{B}_{\text{det}} = \{0, g\} \cup \mathcal{C}$ and every P_c are generated by the same rule on estimation, calibration, and online windows. Under the idealized conditions of a frozen full annihilator, known covariance, and Gaussian noise, each matched energy is the profile GLR for a mean shift in its fixed subspace [39]; these optimality statements do not extend to partial annihilation, data-dependent modes, or the dynamically normalized max. In deployment, every branch

uses the complete deterministic radius $\gamma_{b,k}^{\text{anc}} + \gamma_{b,k}^{\text{det}}$ and the shared calibration layer of Section III-C. Adding a branch can improve directional sensitivity but can also inflate the calibrated maximum; no no-worse power claim is made without recalibration.

C. Input-Scheduled Dynamic Threshold With a Distribution-Free Guarantee

The evaluation of the post-filtered statistics separates two responsibilities that the literature usually conflates. A deterministic *threshold generator* propagates normal-data-estimated envelopes of the structured nuisance terms through the actual window operators, producing an online threshold that follows the measured input, the operating regime, and the reference age; a *conformal layer* then calibrates the observable excess over this deterministic part on held-out normal blocks. The division of labor is stated once and precisely: the false-alarm guarantee of Theorem 1 rests on exchangeability alone and survives misspecification of every envelope; the envelopes are accountable only for detection power and for the detectability statement of Corollary 1, where their validity at fault-time inputs enters as an explicit assumption. The deterministic part is thus a normalization surface with physical meaning—it is what makes the threshold input-scheduled—while the distribution-free layer owns the guarantee.

1) *Input-dependent envelopes*: On the estimation block, the mismatch magnitude $\|\varepsilon_j^m\|$ is regressed against the features $(1, \|u_j\|, \|\Delta u_j\|, \|\Delta \xi_j\|)$ by a nonnegative quantile fit within each fuzzy region c , giving the online envelope

$$\bar{\mu}_j = \eta_0(c_j) + \eta_u(c_j)\|u_j\| + \eta_{\Delta u}(c_j)\|\Delta u_j\| + \eta_\xi(c_j)\|\Delta \xi_j\|, \quad (23)$$

so the measured instantaneous input enters the threshold with a known fitted slope. The context-drift magnitude is enveloped by an age curve $\bar{\delta}(h)$ indexed by the anchor age h , estimated on the same block. Both are empirical normal-data surfaces. Their latent componentwise exceedances are not observable online; only the aggregate observable excess score and descriptor coverage are monitored.

2) *Directional propagation over the stacked window*: Let E_j inject the j th forcing block into the stack and define $C_{b,j} = L_b \mathbf{G}_\nu E_j$ for every $b \in \mathcal{B}_{\text{det}}$. The deterministic threshold is

$$\gamma_{b,k,N}^{\text{det}} = \sum_{j=1}^N (\bar{\mu}_j + \bar{\delta}(h_j)) \|C_{b,j}\|, \quad (24)$$

computed with an exact SVD or a certified spectral-norm upper bound. Uncertified finite power iteration is used only as a lower-bound witness for the aggregation-slack diagnostic, never in the sound threshold. When (16) is nondegenerate, each displayed block norm means the certified supremum over the jointly propagated operator enclosure; the same operator-error element is not independently maximized in different physical paths when a tighter joint support-function calculation is used.

Proposition 5 (Containment for the first-order surrogate). *If $\|\varepsilon_j^m + d_j^x\| \leq \bar{\mu}_j + \bar{\delta}(h_j)$ for all window steps j , then every branch satisfies $\|L_b \mathbf{G}_\nu(\varepsilon^m + d^x)\| \leq \gamma_{b,k,N}^{\text{det}}$.*

The proof is the triangle inequality over block columns. It is sound for the declared surrogate but not the exact radius of the source sum. For independent compact source sets M_j , the exact deployed-operator radius is

$$\gamma_{b,k,N}^{\text{ex}} = \max_{\|u\|=1} \sum_j h_{M_j}(C_{b,j}^\top u); \quad (25)$$

for correlated sources, the sum must be replaced by the support function of their joint feasible set. Equation (24) preserves direction within each source and combines sources worst-case by the triangle inequality. The complete branch radius is $\gamma_{b,k}^{\text{anc}} + \gamma_{b,k}^{\text{det}}$; the bilinear remainder is not covered by either term and is priced by the calibration layer below.

Remark 1 (Honest comparison with scalar recursions). The bound (24) is not pointwise tighter than a scalar-radius recursion: summing block-column norms is worst-case across sources, and in fast-mixing regimes a root-sum-of-squares recursion can be smaller. Its advantages are branch-specific evaluation through the actual L_b and explicit separation of the retained anchor radius (21). The aggregation-slack ratio between a certified upper bound and a positive support-function witness is reported per window; it is undefined, and reported as N.A., when the propagated nuisance radius is zero.

3) *Conformal layer and the false-alarm guarantee*: For each simultaneously computed branch b , define the observable excess score $\tau_{b,k} = \max\{0, T_{b,k} - \gamma_{b,k}^{\text{anc}} - \gamma_{b,k}^{\text{det}}\}/\sigma_b$, where each $\sigma_b > 0$ is frozen on the estimation block; a zero-scale branch is disabled or assigned a prespecified positive floor before calibration. Let $\tau_k = \max_{b \in \mathcal{B}_{\text{det}}} \tau_{b,k}$. The post-filter candidate, active-mode rules, windows, and detection score map are frozen before final calibration. On a prespecified finite monitoring episode, one maximum is taken over all selectable times, modes, and detection branches. Each episode starts from the same deterministic monitor reset state, or includes the complete finite monitor state and its burn-in as part of the exchangeable unit; a maximum anchor age makes that memory finite. The online branch threshold is

$$\gamma_{b,k}^{\text{dyn}} = \max\{\epsilon_\Gamma, \gamma_{b,k}^{\text{anc}} + \gamma_{b,k}^{\text{det}} + \sigma_b q_{1-\alpha}^{\text{det}}\}, \quad (26)$$

and write $\Gamma_{b,k} := \gamma_{b,k}^{\text{dyn}}$. Here $\epsilon_\Gamma > 0$ is frozen before calibration and $q_{1-\alpha}^{\text{det}}$ is the conformal rank quantile of the frozen detection-episode maximum. The positive floor makes every deployed gauge finite and can only make the alarm more conservative. For any frozen episode-score family with n_{cal} calibration episodes and error budget δ , this paper uses the finite-sample rule

$$j_\delta = \lceil (n_{\text{cal}} + 1)(1 - \delta) \rceil, \quad (27)$$

$$q_{1-\delta} = \begin{cases} S_{(j_\delta)}, & j_\delta \leq n_{\text{cal}}, \\ +\infty, & j_\delta = n_{\text{cal}} + 1, \end{cases}$$

with a strict exceedance decision and no anti-conservative tie breaking. A finite threshold requires $\delta \geq 1/(n_{\text{cal}} + 1)$; otherwise the corresponding decision is disabled. Mondrian cells may still be used inside the deterministic normalization, but separate marginal cell quantiles do not by themselves protect adaptive cell selection.

Theorem 1 (Finite-episode simultaneous false-alarm guarantee). *Suppose (i) training, candidate design, and final calibration use disjoint normal blocks, and the complete detection score map together with every score-generating monitor transition—including the anchor gate—is frozen before calibration; (ii) the calibration episode-maximum scores and the target normal-episode maximum score are exchangeable; and (iii) every selectable time, mode, window, and detection branch is included in that maximum, while the final quantile never feeds back into their state path. Then, in fault-free operation over the prespecified target episode,*

$$\Pr \left\{ \max_{k \text{ in episode}} \tau_k > q_{1-\alpha}^{\text{det}} \right\} \leq \alpha, \quad (28)$$

and every alarm stopping time within that episode inherits the same detection event. This theorem does not calibrate the post-alarm explanation sets, mask outputs, or fault-time signature coverage.

The proof, in the Appendix, is the split-conformal rank argument [27] applied to the frozen superblock-maximum score. This finite-episode statement must not be iterated indefinitely: selecting the first violating block over an unbounded sequence is not protected by a per-block marginal level. A declared finite collection of episodes requires one joint maximum or an explicit error-budget allocation; a genuinely indefinite guarantee would require an always-valid construction not developed here. Approximate block laws require an additional dependence bound rather than the claim that block reduction itself creates exchangeability. Because marginal coverage does not imply conditional coverage [41], the protocol also reports regime-wise empirical performance without presenting it as a conditional theorem.

4) *Class-conditional detectability*:

Corollary 1 (Minimum detectable persistent fault). *In the fixed linear special case, let the class branch use the same deployed operator $L_c = P_c L_0$ in its statistic, threshold, and signature, and set $A_c = L_c F_c (\mathbf{1}_N \otimes I)$. For the declared positively homogeneous parameter set Θ_c , define the restricted minimum gain $\underline{s}_c = \inf\{\|A_c f\| : f \in \Theta_c, \|f\| = 1\}$. Let $\mathbf{f} = \mathbf{1}_N \otimes f_0$ with $f_0 \in \Theta_c$, assume $\underline{s}_c > 0$, and assume the independent fault-time nuisance event of Assumption A5' holds. If*

$$\underline{s}_c \|f_0\| > 2\gamma_{c,k}^{\text{dyn}}, \quad (29)$$

then $T_{c,k} > \gamma_{c,k}^{\text{dyn}}$ on that window. This is a deterministic statement conditional on a fault-time nuisance event; Theorem 1 does not assign that event an H_1 probability.

The positivity of $\underline{s}_c$ is an injectivity requirement on the allowed fault directions; for the full parameter space it is equivalent to full column rank. If it fails, only a direction-specific condition $\|A_c f_0\| > 2\gamma_{c,k}^{\text{dyn}}$ is valid. No sliding-window or m -of- n conclusion follows from a single-window inequality.

Remark 2 (Adaptive blindness is observable, not absent). A fault that changes the recorded input may inflate γ^{det} through the known slopes of (23). This yields only a per-window, path-conditional stealth set. It does not bound total energy

independently of duration: a nonzero signature that remains below every window threshold can accumulate unbounded energy over an indefinite run. Any reachable $+\infty$ calibration cell or guard-only mode also destroys a finite omnibus ceiling. The input-driven and mode-dependent threshold contributions are therefore logged at every alarm and suppressed alarm.

D. Path-Conditioned Set-Valued Isolation

Isolation reuses the frozen alarm window unless a persistence assumption supplies an explicit map to a later window. It conditions only on trusted side information z_k^{safe} —the exogenous trajectory and scheduling variables proved insensitive to the fault. A clean anchor is a separate signature-coverage event, not an observed coordinate: an undetected persistent sensor bias established before the anchor can be inherited by both branches and make even the guard residual vanish. Fault-sensitive memberships and gates remain latent variables in the explanation set; conditioning on their observed faulty point estimate could exclude the true class.

1) *Physical explanation families*: For class c , let $\mathcal{A}_c(k \mid z_k^{\text{safe}})$ be a compact joint admissible graph containing the operating path, onset, temporal profile, fault parameter, model uncertainty, and hybrid mode. A single element a simultaneously determines the signature $v_c(a)$, nuisance set $\mathbb{U}_c(a)$, active linear branches, thresholds, and mask-conditioned recomputations. The actuator family is generated by the learned input response only on operating directions supported by the normal conditional input-excitation Gramian; without such excitation or external physical knowledge, it is labeled a *learned-model input-side explanation*, not a confirmed physical actuator fault. The sensor encoder-JVP family has the analogous boundary: normal histories may lie on a lower-dimensional manifold, so transverse measurement derivatives require external calibration, perturbation coverage, or a verified structural constraint. The process family requires an externally specified injection map or uncertainty family, because normal data with $f^p = 0$ contain no information that identifies that map. An unrestricted per-step process forcing fills the retained residual space and makes actuator–process isolation impossible.

There is a further fault–nuisance gauge equivalence already at the raw-data level. A sensor fault can be absorbed into measurement noise; a process fault can be absorbed into process disturbance; and the state increment induced by an actuator fault can be absorbed into process disturbance. Hence the raw record identifies only the total measurement- and dynamics-side injections unless bounded-support separation, persistence, or external physics supplies an operational fault definition. Coincident full topological supports rule out a label that is correct at every support point; an almost-sure zero-error impossibility additionally requires overlapping positive density or mutual absolute continuity, as in finite-dimensional Gaussian models. The sets below therefore use declared high-probability nuisance tubes.

The sensor- j family uses one joint map for the encoder, current-output, and recorded-command paths,

$$v_{s-j}(a) = [F_{\text{enc},j}(a) + F_{\text{direct},j}(a) + F_{\text{cmd},j}(a)]\theta_j. \quad (30)$$

The three marginal images are not added independently because they share the same fault and operating path. The counterfactual command response required to construct $F_{\text{cmd},j}$ is physically identifiable only when the controller law or a certified local response map from measurement perturbations to commands is available. Otherwise, $F_{\text{cmd},j}$ must range over an externally declared uncertainty family, and the result is only a sensor explanation conditional on the actually recorded command sequence, not an unconditional physical sensor signature. Recomputing the entire pipeline under a channel mask is a counterfactual nonlinear observation, not generally a linear map of the unmasked stacked residual. Accordingly, define the deployed observation

$$\mathcal{O}_k^{\text{full}}(a, n) = \Phi(m_{k-}(a, n), D_k(a, n)) = (m_{k-}, D_k, z_k^{\text{safe}}, \mathcal{B}_{\text{act}}, \{L_{b,k}, T_{b,k}\}), \quad (31)$$

where $D_k = (u, y, \xi)_{\text{window}}$, m_{k-} is the complete finite monitor state immediately before the decision, and Ψ_k contains every unmasked/masked recomputation and r^y . This pair (m_{k-}, D_k) determines the anchor age, modes, operator enclosures, branches, statistics, thresholds, and oracle lifecycle; listing them in (31) makes that sufficiency auditable. A true regime, fault amplitude, class-specific oracle threshold, or unobserved mode is not an admissible observation coordinate. Consequently, if two physical explanation sets intersect in (m_{k-}, D_k) , no mask, encoder, or post-filter can make them physically identifiable; the deterministic map Φ may only preserve or lose information. Algorithmic memberships, selectors, and active modes may be included in $\mathcal{O}^{\text{full}}$ only when they are frozen measurable functions of (m_{k-}, D_k) ; identical explanations in that space necessarily receive the same values. By contrast, a healthy counterfactual regime or fault-conditioned physical mode is latent and must remain inside $\mathcal{A}_c$ or a certified set-valued cover. The associated class explanation set is

$$\mathbb{E}_c^{\text{full}}(k \mid z_k^{\text{safe}}) = \{\mathcal{O}_k^{\text{full}}(a, n) : a \in \mathcal{A}_c(k \mid z_k^{\text{safe}}), n \in \mathbb{U}_c(a)\}. \quad (32)$$

For computation, each class has a frozen Borel outer family on $q \in [0, +\infty]$ satisfying $\overline{\mathbb{E}}_c(q_1) \subseteq \overline{\mathbb{E}}_c(q_2)$ for $q_1 \leq q_2$, $\overline{\mathbb{E}}_c(+\infty)$ equal to the declared observation universe, and the right-closure condition $\overline{\mathbb{E}}_c(q) = \bigcap_{r>q} \overline{\mathbb{E}}_c(r)$. Hence the measurable score

$$s_c(\mathcal{O}) = \inf\{q : \mathcal{O} \in \overline{\mathbb{E}}_c(q)\} \quad (33)$$

obeys $\overline{\mathbb{E}}_c(q) = \{\mathcal{O} : s_c(\mathcal{O}) \leq q\}$, including at boundaries. Its cell-refinement hierarchy and cache are shared across q ; unresolved cells remain feasible, so a larger q cannot become computationally less feasible merely because it received a different refinement budget. For fault classes, the deployed radius q_c comes from the declared physical/operator and fault-time nuisance bounds; normal calibration does not determine it.

The normal attribution set must inspect the same full observation as the fault classes. The normal calibration data are pre-partitioned into disjoint detection- and attribution-calibration episodes. First $q_{1-\alpha}^{\text{det}}$ is computed and frozen on the detection subset. Only then, on every attribution-calibration

episode, the frozen mask/enclosure/oracle pipeline is replayed at every selectable hypothetical isolation time, and $S_i^0 = \max_t s_0(\mathcal{O}_{i,t}^{\text{full}})$ is calibrated by (27) at budget β . This ordering is necessary because $\mathcal{O}^{\text{full}}$ contains the deployed Γ_b , which depends on the already-frozen detection quantile. With $0 < \beta < \alpha$, define

$$\mathbb{A}_0^{\text{det}} = \{\mathcal{O}^{\text{full}} : \tau^{\text{det}} \leq q_{1-\alpha}^{\text{det}}\} \subseteq \mathbb{E}_0^{\text{attr}} = \mathbb{A}_0^{\text{det}} \cup \mathbb{E}_0^{\text{full}}(q_{1-\beta}^{\text{attr}}). \quad (34)$$

Thus an alarm can remain NORMAL-COMPATIBLE, but only when its complete counterfactual observation has a calibrated normal explanation. If $\beta < 1/(n_{\text{cal}}^{\text{attr}} + 1)$, then $q_{1-\beta}^{\text{attr}} = +\infty$, the normal class cannot be excluded, and nonnormal singleton reporting is disabled.

2) *Set-valued decision and its linear special case:* Given the observed tuple $\mathcal{O}_{\text{obs}}^{\text{full}}$, the output is

$$\mathcal{C}_k^+(\mathcal{O}_{\text{obs}}^{\text{full}}) = \{0 : \mathcal{O}_{\text{obs}}^{\text{full}} \in \mathbb{E}_0^{\text{attr}}\} \cup \{c \in \mathcal{C}_{\text{fault}} : \mathcal{O}_{\text{obs}}^{\text{full}} \in \mathbb{E}_c(q_c)\}, \quad (35)$$

where $c = 0$ is the nuisance-only normal explanation. An alarm is not a fault-presence certificate because the detector has nonzero false-alarm risk.

Theorem 2 (Coverage-conditional isolation and family-wise normal risk). *Let $c^* = (c_t^*)_{t \in \mathcal{T}_{\text{sel}}}$ be the true class/support trajectory on the finite set of every selectable report time/mode/window/support, with $c_t^* = 0$ before onset and piecewise changes allowed for sequential or mixed faults. Let $\mathcal{E}_{\text{sig}, c^*}^{\text{ep}} = \bigcap_{t \in \mathcal{T}_{\text{sel}}} \mathcal{E}_{\text{sig}, c_t^*, t}^{\text{ep}}$ and $\mathcal{E}_{\text{nuis}, c^*}^{\text{ep}} = \bigcap_{t \in \mathcal{T}_{\text{sel}}} \mathcal{E}_{\text{nuis}, c_t^*, t}^{\text{ep}}$, where the same joint path generates all time-indexed explanations. On their intersection, $c_t^* \in \mathcal{C}_t^+$ at every selectable time. If additionally $\Pr_{c^*}((\mathcal{E}_{\text{sig}, c^*}^{\text{ep}})^c) \leq \delta_{c^*}$ and $\Pr_{c^*}((\mathcal{E}_{\text{nuis}, c^*}^{\text{ep}})^c) \leq \epsilon_{c^*}$, the probability that there exists a reported singleton $\mathcal{C}_t^+ = \{c\}$ with $c \neq c_t^*$ is at most $\delta_{c^*} + \epsilon_{c^*}$. Under H_0 , episode calibration of (34) gives $\Pr_0\{\text{any nonnormal singleton in the episode}\} \leq \beta$, because such an output must exclude the normal attribution set. A set containing 0 is NORMAL-COMPATIBLE, an empty set is OUT-OF-MODEL, and multiple classes are NONUNIQUE. Pointwise H_1 coverage is insufficient for the displayed stopped/selected episode risk. No per-label posterior confidence, FDR, PPV, or H_1 class risk follows from the detection level α or the normal budget β . When an episode-level δ_c or ϵ_c is unavailable, the corresponding statement remains simultaneous-coverage-conditional only.*

For the sublibrary of simultaneously computed linear branches derived from one frozen residual, define

$$\|x\|_{*,k} = \max_{b \in \mathcal{B}_{\text{act}}} \frac{\|L_{b,k}x\|}{\Gamma_{b,k}}. \quad (36)$$

After quotienting its joint kernel, this is a norm when all active Γ_b are finite and positive. In this special case define $\mathbb{K}_c(k | z_k^{\text{safe}}) = \{v_c(a) : a \in \mathcal{A}_c(k | z_k^{\text{safe}})\}$ and suppose class c has a complete symmetric nuisance ball $\mathbb{U}_c = \{n : \|n\|_{*,k} \leq r_c\}$. A true signature v from c^* excludes every competing fault class whenever

$$\min_{c \in \mathcal{C}_{\text{fault}}, c \neq c^*} \text{dist}_{*,k}(v, \mathbb{K}_c(k | z_k^{\text{safe}})) > r_{c^*} + r_c. \quad (37)$$

If all fault-class radii equal one, the constant two is exact for that complete common-ball model when the competing distance is attained. The normal competitor is different: it is the full-observation set $\mathbb{E}_0^{\text{attr}}$ in (34), so it must be excluded by direct explanation-set disjointness. Only when its relevant residual slice is representable as $\mathbb{K}_0 \oplus \{n : \|n\|_{*,k} \leq r_0^{\text{attr}}\}$ does this reduce to $\text{dist}_{*,k}(v, \mathbb{K}_0) > r_{c^*} + r_0^{\text{attr}}$. Using the detection radius one for that attribution set is invalid. Fixed-subspace principal-angle bounds are conservative computable corollaries of the fault-fault part of (37), not the general decision rule.

3) *Mixed faults and reporting boundary:* For a support S , the mixed family is generated by a joint nonlinear map $g_S(a, \{\theta_c\}_{c \in S})$. The sum of single-fault signatures is only a local first-order approximation; its cross-fault remainder is bounded and included in a mixed-specific nuisance set. A component c is reported as necessarily present only if every feasible support not containing c is excluded. The full support dictionary is exponential, so the implemented restriction to at most one sensor fault plus one dynamics-side fault is declared as a concurrency prior. If a physical actuator map or process injection map is not identifiable under the stated information set, the output is the corresponding learned-model or process-side equivalence class rather than a physical confirmation.

Masking remains useful because it removes the raw channel from all encoder-mediated paths when the complete pipeline is recomputed, but the command-dependent model-error path is generally first order and the nonlinear recomputation remainder is explicitly budgeted. Failure of masked exoneration therefore never excludes a sensor explanation. A premise failure triggers fail-closed output only when an implemented monitor actually detects it. An unobserved failure of $\mathcal{E}_{\text{sig}}$ may instead yield a stable but wrong singleton; this is why signature coverage is reported as an unverified physical assumption wherever no independent bound exists.

E. Learning and Calibration Protocol

All model components are trained jointly on the normal training block. The history (8) is tokenized per instant, embedded linearly, and processed by a causal multi-head attention encoder with smooth activations, so that every Jacobian used above exists and is continuous; the premise network is a small bottleneck MLP; the rule metrics use a bounded-spectrum parameterization; the local matrices A_i carry a soft spectral cap as a numerical safeguard; and the output map C_z is linear, which the isolation layer exploits. Training minimizes a free-running multi-horizon loss: anchors s and horizons $N \sim \text{Unif}\{1, \dots, N_{\text{max}}\}$ are sampled, the same rollout (10) used online produces $\hat{z}_{s+j|s}$ and $\hat{y}_{s+j|s}$ without reading intermediate measurements, and weighted latent and output prediction errors are accumulated, together with a decoding-consistency term and a variance regularizer that jointly exclude latent collapse, a membership-balance term that keeps all rules in use without forbidding sharp partitions, and a finite-horizon product penalty on the complete Jacobian norms that discourages explosive rollouts. Channel masking

is applied during training with a fixed distribution (all-pass, single-channel, and independent multi-channel masking), which trains the shared mask-conditioned encoder used for isolation without degrading the unmasked task.

Calibration discipline is stricter than usual and is essential for the guarantee of Section III-C: the normal data are split into *five* disjoint frozen blocks—(i) a training block for all learnable parameters; (ii) an estimation block on which the mismatch envelopes, the context-drift curve, stacked covariances, branch scales, positive threshold floor, anchor-eligibility gate, complete mode/state transitions, and outer-family refinement hierarchy are fixed; (iii) a detection-calibration block on which only $q_{1-\alpha}^{\text{det}}$ is computed; (iv) an attribution-calibration block on which that detection quantile is treated as fixed and only $q_{1-\beta}^{\text{attr}}$ is computed from full-normal scores; and (v) a held-out normal test segment that joins the frozen fault test set. Fitted preprocessing is learned on the training block only, and isolation gaps of length $\ell_h + N_{\max}$ separate consecutive blocks. The frozen re-anchoring policy is replayed identically on every calibration episode, but neither quantile may feed back into the state path that generated its own calibration scores, and the detection-calibration episodes are not reused for attribution calibration. No fault data influence any of these steps.

F. Online Operation and Complexity

The set-membership oracle is fixed before calibration. For each declared class/support and finite hybrid mode, its compact continuous parameter domain is partitioned by one shared refinement tree. Analytic, interval, or certified spectral bounds map each cell to a nested outer enclosure of the complete observation (31), including numerical tolerance. A class is feasible if at least one enclosure intersects the observed tuple and its synchronized tube. Cells for competing explanations are refined along the frozen hierarchy until they separate or the declared alarm-time budget is exhausted; unresolved competitors remain in the candidate set, and the common monotone cache enforces the right-closed nesting required by (33). Thus numerical approximation can increase NONUNIQUE but cannot unsafely exclude a declared explanation. The normal class $c = 0$ is always included in the dictionary and its full oracle is replayed at every hypothetical calibration-time report. Mixed supports are restricted by the concurrency prior stated above.

Before an alarm, the costs are the encoder/rollout, the JVPs required by every detection branch, certified norm bounds, and the thin decompositions that update W_0 and Q_w under the frozen candidate rule. These same operations are replayed during calibration. Alarm-time isolation additionally costs m_y full mask-conditioned recomputations and a hybrid-cell membership search. Its worst-case cost is exponential in the allowed support size and in the continuous parameter dimension; the stated concurrency prior and an explicit refinement budget make it finite, but no general real-time claim is made for the isolation solver. Runtime and unresolved-cell rates are reported separately from the detection-front-end latency.

Algorithm 1 Protected monitoring at instant k

- 1: encode $d_k^- \rightarrow (z_k^E, \chi_k^E)$; advance short- and long-term rollouts (10)
 - 2: update ν_{k-1} , $\delta_{k-1|s}^x$, r_k^y , and the window stack $e_{s,N}$
 - 3: assemble/update certified enclosures of G_ν, G_0 , then W_0, K_{anc}, Q_w and every calibrated detection-branch operator L_b
 - 4: compute all detection statistics and their complete thresholds with the same L_b
 - 5: update the pre-calibration-frozen anchor gate and confirm/advance anchors independently of the final quantiles
 - 6: emit the final detection alarm without feeding it back into the score-generating monitor path
 - 7: **if** alarm **then**
 - 8: recompute all declared masks and form $\mathcal{O}_{\text{obs}}^{\text{full}}$, query the certified outer-cell oracle
 - 9: report $\mathcal{C}_k^+$ as fault singleton, NORMAL-COMPATIBLE, NONUNIQUE, equivalence class, or OUT-OF-MODEL
 - 10: **end if**
-

IV. EXPERIMENTAL PROTOCOL

The method targets closed-loop process data under the only-normal-data protocol of Assumption 2. This section fixes the complete evaluation protocol. In accordance with that protocol, all numerical entries are produced by a single frozen evaluation run executed after every design quantity of Sections III–III-F has been frozen; the result placeholders below are intentionally left open in this manuscript version and no number reported here was obtained from fault data.

A. Benchmark and Data Split

The primary benchmark is a closed-loop continuous stirred-tank reactor (CSTR) simulation dataset with commands $u = (C_i, T_i, T_{ci})$ and measured outputs $y = (C, T, T_c, Q_c)$, one long normal record, and eight faulty records with a common fault onset; faults 1–2 are process faults, 3–5 actuator faults, and 6–8 sensor faults, so the benchmark covers exactly the three structured classes of Section III-D. The normal record is partitioned 55/15/10/10/10% into the five blocks of Section III-E with isolation gaps; the faulty records join the frozen test set with per-sample labels beginning at the true onset.

B. Numerical Design Verification

Independent randomized checks currently verify only algebraic subcomponents: the stacked propagation identity, the full-annihilator equality $Q^T G_0 = 0$, the predicted first-order/bilinear scaling, the banded encoder response, and containment of sampled nuisance realizations by a certified block-norm sum. They do not validate the new white-space partial quotient, physical signature coverage, the hybrid outer-cell oracle, or Theorem 2; those require the explicit tests below and remain unfilled in this draft. In particular, no principal-angle classification count is used as evidence for the set-valued theorem.

TABLE I

DETECTION RESULTS ON THE FROZEN CSTR TEST SET (TO BE COMPLETED BY THE FROZEN EVALUATION RUN; NO ENTRY MAY BE FILLED FROM ANY RUN THAT INFLUENCED DESIGN).

Method	FAR	MDR	Delay	ARL ₀
PCA T^2 /SPE	–	–	–	–
Reconstruction AE	–	–	–	–
One-step predictor	–	–	–	–
Koopman parity [16]	–	–	–	–
Proposed (full)	–	–	–	–
Ablations A1–A5	–	–	–	–

TABLE II

ISOLATION RESULTS ON THE FROZEN CSTR TEST SET (TO BE COMPLETED BY THE FROZEN EVALUATION RUN; REJECTION IS REPORTED AS AN EXPLICIT OUTCOME).

Method	Fault singleton	Normal-compatible	Nonunique	Out-of-model
Proposed	–	–	–	–
A6 (no masking)	–	–	–	–
A7 (forced label)	–	–	–	–

C. Baselines and Ablations

Baselines span the three traditions of Section I: static multivariate monitoring (PCA with T^2 /SPE, a reconstruction autoencoder), a one-step dynamic predictor with a constant calibrated threshold, and a measurement-driven Koopman parity design representative of [15], [16]. Ablations remove one design element at a time: (A1) no anchor annihilation; (A2) no matched branches (omnibus only); (A3) scalar-radius threshold in place of the block-column generator; (A4) conformal calibration of the raw score without the deterministic part; (A5) envelope without input features; (A6) no counterfactual mask recomputations; and (A7) forced singleton output without set-valued rejection. Each ablation isolates one claim of Sections III-B–III-D.

D. Metrics

Detection is scored by finite-episode family-wise false-alarm rate, missed-detection rate, delay, and empirical run length. Threshold reporting separates γ^{anc} , γ^{det} , the calibration term, and ϵ_Γ , together with $n_{\text{cal}}^{\text{det}}$, $n_{\text{cal}}^{\text{attr}}$, attainable α/β resolution, monitor reset state, and unresolved/infinite modes. Isolation reports the frequencies of nonnormal singleton, NORMAL-COMPATIBLE, NONUNIQUE, equivalence-class, and OUT-OF-MODEL outputs; conditional correctness is reported only among nonnormal singletons. Pointwise fault-time coverage diagnostics are reported separately from the simultaneous selected-episode $\mathcal{E}_{\text{sig}}^{\text{ep}}/\mathcal{E}_{\text{nuis}}^{\text{ep}}$ assumptions used by Theorem 2. Additional diagnostics are full-state/raw-record class overlap, clean-anchor ambiguity, conditional input-excitation rank, sensor transverse-JVP sensitivity, process-map prior width, mixed-interaction budget, outer-cell refinement count, monotonicity violations, and unresolved-support rate.

V. CONCLUSION

This paper turned the informal requirement “learn normal behavior and monitor against it” into a protected architecture with three explicitly separated layers. Conditional on the recorded commands, the protected dual-branch design removes direct post-anchor measurement correction while retaining the closed-loop measurement–command path, and the exact propagation identity reduces monitoring to the structure of one stacked residual. On that structure, the energy-selective anchor quotient with a guard branch exposes and prices its retained nuisance and blind directions; the input- and path-scheduled threshold separates deterministic set containment from a finite-episode simultaneous calibration statement; and the isolation layer replaces forced attribution by operating-path-conditioned explanation sets that include the mask-conditioned counterfactual outputs and return equivalence classes when physical structure is insufficient.

The limitations are structural. A recorded command common to both branches cancels, whereas unrecorded actuator and process injections do not; their physical labels nevertheless require input excitation and an externally specified process map that only-normal data cannot supply. Partial anchor annihilation leaves a priced residual, and a fault aligned with the removed subspace remains a guard-only case. Normal calibration controls neither fault-time nuisance coverage nor signature-model coverage. A per-window threshold does not bound indefinite accumulated stealth energy, and finite-episode block calibration does not yield an always-valid first-alarm guarantee. Completion of the frozen benchmark evaluation, physical validation of the declared signature families, and an always-valid extension for indefinite monitoring are therefore separate next steps.

APPENDIX

A. Complete Fuzzy Jacobians and Dispersion Bounds

Proposition 6 (Complete T–S Koopman Jacobians). *Let the memberships be the softmax of scores $\varsigma_i(\rho) = -\frac{1}{2}(\rho - p_i)^T M_i(\rho - p_i) + o_i^\omega$ with $M_i = M_i^T \succ 0$, and let $\mathcal{P}$ be continuously differentiable. With $\varsigma_i^\nabla = -M_i(\rho - p_i)$, $\bar{\varsigma}^\nabla = \sum_i \omega_i \varsigma_i^\nabla$, $\bar{v} = \sum_i \omega_i v_i$, and $v_i = A_i z + B_i u + o_i$,*

$$\begin{aligned} \nabla_\rho \omega_i &= \omega_i (\varsigma_i^\nabla - \bar{\varsigma}^\nabla), \\ J_z \mathcal{Z}_\Theta &= A_\omega + \Delta_\rho J_{\rho,z}, \quad J_u \mathcal{Z}_\Theta = B_\omega + \Delta_\rho J_{\rho,u}, \\ \Delta_\rho &= \sum_i \omega_i (v_i - \bar{v}) (\varsigma_i^\nabla - \bar{\varsigma}^\nabla)^T, \end{aligned} \quad (38)$$

with $A_\omega = \sum_i \omega_i A_i$ and $B_\omega = \sum_i \omega_i B_i$, and the dispersion bound

$$\|\Delta_\rho J_{\rho,z}\| \leq \sqrt{\lambda_{\max}(\Sigma_v)} \sqrt{\lambda_{\max}(J_{\rho,z}^T \Sigma_\varsigma J_{\rho,z})}, \quad (39)$$

where $\Sigma_v = \sum_i \omega_i (v_i - \bar{v})(v_i - \bar{v})^T$ and $\Sigma_\varsigma = \sum_i \omega_i (\varsigma_i^\nabla - \bar{\varsigma}^\nabla)(\varsigma_i^\nabla - \bar{\varsigma}^\nabla)^T$.

Proof. The softmax gradient follows from logarithmic differentiation. Differentiating $\mathcal{Z}_\Theta = \sum_i \omega_i v_i$ in z gives A_ω plus $\sum_i v_i (\nabla_\rho \omega_i)^T J_{\rho,z}$; inserting the gradient identity and using $\sum_i \omega_i (\varsigma_i^\nabla - \bar{\varsigma}^\nabla) = 0$ centers v_i to $v_i - \bar{v}$, giving Δ_ρ . Writing $\Delta_\rho = V_w S_w^T$ with $V_w = [\sqrt{\omega_i}(v_i - \bar{v})]_i$ and $S_w =$

$[\sqrt{\omega_i}(\zeta_i^\nabla - \bar{\zeta}^\nabla)]_i$ and applying the submultiplicative bound $\|V_w S_w^\top J_{\rho,z}\| \leq \|V_w\| \|S_w^\top J_{\rho,z}\|$ yields (39). All expressions were verified against central differences to 10^{-10} . $\square$

The second term of (38) vanishes as the memberships approach one-hot and can dominate near rule boundaries; estimating the propagation gain by $\|A_w\|$ alone therefore systematically underestimates it. The same structure bounds the context sensitivity $\|J_\chi \mathcal{Z}_\Theta\|$, which enters the drift envelope $\bar{\delta}(h)$ and the buffer path of the sensor signature.

B. Proof of Proposition 2

By the fundamental theorem of calculus along the segment from $\hat{z}_{k|s}$ to z_k^E with the protected context frozen, $\mathcal{Z}_\Theta(z_k^E, u_k, \xi_k; \chi_{k|s}^n) - \mathcal{Z}_\Theta(\hat{z}_{k|s}, u_k, \xi_k; \chi_{k|s}^n) = A_{k|s} e_{k|s}$ with $A_{k|s}$ as in (13). Adding and subtracting $\mathcal{Z}_\Theta(z_k^E, u_k, \xi_k; \chi_{k|s}^E)$ and using (12) gives the one-step recursion $e_{k+1|s} = A_{k|s} e_{k|s} + \nu_k + \delta_{k|s}^\chi$; no fault signal is differentiated because the identity manipulates definitions only. Unrolling from $e_{s|s} = 0$ yields (14), and stacking rows gives (15). $\blacksquare$

C. Proof Sketch of Proposition 3

Insert $z^E = z^* + \varepsilon^e$ into the combined forcing of Lemma 1 and expand to first order at z^* : the forcing splits into the ideal-trajectory mismatch and noise $\varepsilon_k^m + \varepsilon_k^v$, the context-drift term d_k^χ , the encoder-noise difference $\varepsilon_{k+1}^e - J_z \mathcal{Z}_\Theta(z_k^*) \varepsilon_k^e$, and a Taylor remainder quadratic in ε^e . Propagating through (15) and telescoping $\sum_j \Phi(l, j+1)(\varepsilon_{j+1}^e - A_j \varepsilon_j^e) = \varepsilon_l^e - \Phi(l, s) \varepsilon_s^e + \rho_l$ leaves the telescoping defect ρ_l , which collects the products $\Phi(l, j+1)(A_{j|s} - J_z \mathcal{Z}_\Theta(z_j^*)) \varepsilon_j^e$; since the segment-averaged and pointwise Jacobians differ by at most $L_H(\|e_{j|s}\| + \|\varepsilon_j^e\|)$, the defect and the Taylor remainder together obey the stated bilinear bound. The signature terms (18) follow by the same first-order expansion of the faulty data branch: actuator faults perturb the applied input at each step and propagate through G_ν ; sensor faults perturb the encoder inputs, whose effect telescopes to the direct banded image (plus the buffer path when the fault predates the anchor); process faults perturb the state channel and propagate through G_ν with unknown per-step injection. The numerical scaling study of Section IV-B confirms both the clean first-order rates and the bilinear floor. $\blacksquare$

D. Assumption A5' and Proofs for Section III-C

Assumption A5' (fault-time nuisance event). Along the fault-perturbed trajectory up to the decision instant, the fault-free forcing components satisfy $\|\varepsilon_j^m + d_j^\chi\| \leq \bar{\mu}_j + \bar{\delta}(h_j)$ evaluated at the realized inputs and regimes, and all other fault-time nuisance components lie in the declared tube. This is an independent H_1 assumption about model validity off the normal input distribution; Theorem 1 assigns it no probability.

Proof of Theorem 1. Once the complete detection pipeline and its state transition are frozen, each calibration superblock and the target normal episode are mapped measurably to one scalar maximum over every selectable time, detection branch, and detection mode. Post-alarm masks and class-specific explanation scores are not coordinates of this detection

maximum. Under exchangeability, the target maximum has an exchangeable rank among $n_{\text{cal}} + 1$ maxima. The index in (27), together with the strict exceedance convention, therefore gives $\Pr\{\tau^* > q_{1-\alpha}^{\text{det}}\} \leq \alpha$ [27]; when the requested rank is $n_{\text{cal}} + 1$, the $+\infty$ convention is the only distribution-free finite-sample choice. Every online choice within the episode is dominated by τ^* , which proves (28); the positive floor in (26) only shrinks the actual alarm event. The argument does not extend by repeatedly conditioning on the first violating episode. $\blacksquare$

Proof of Corollary 1. On the declared fault-time nuisance event, the branchwise fault-free content is at most $\gamma_{c,k}^{\text{dyn}}$. By the reverse triangle inequality and the definition of the restricted gain, $T_{c,k} \geq \underline{\varepsilon}_c \|f_0\| - \gamma_{c,k}^{\text{dyn}}$. The strict condition (29) therefore gives $T_{c,k} > \gamma_{c,k}^{\text{dyn}}$. $\blacksquare$

E. Proof of Theorem 2

On $\mathcal{E}_{\text{sig}, c^*}^{\text{ep}} \cap \mathcal{E}_{\text{nuis}, c^*}^{\text{ep}}$, the true full observation belongs to the explanation set indexed by c_t^* at every selectable report time, hence c_t^* belongs to every corresponding set in (35). The union bound gives $\Pr_{c^*}\{\exists t : c_t^* \notin C_t^+\} \leq \delta_{c^*} + \epsilon_{c^*}$ whenever both episode budgets exist. If a candidate set is a singleton, every explanation satisfying the declared model has the same time- t class; if it contains two classes, the observed tuple itself is a common feasible observation, so no deterministic single-label rule can be uniformly correct over both explanations. Under H_0 , $c_t^* = 0$ for all t , and a nonnormal singleton requires exclusion of $\mathbb{E}_0^{\text{attr}}$; because the full normal score was maximized over every selectable hypothetical report time before applying (27), its episode probability is at most β . This event argument supplies neither PPV nor a class-conditional H_1 error rate. In the linear class-radius special case at one time t , the true class remains feasible because its distance from $v + n$ is at most $r_{c_t^*}$. A fault competitor is infeasible whenever $\text{dist}_{*,k}(v, \mathbb{K}_c) > r_{c_t^*} + r_c$, by the reverse triangle inequality. If both radii equal one and the competing minimum not exceeding two is attained, the midpoint construction proves sharpness for the complete symmetric-ball model. The full normal attribution set is excluded directly and is not covered by this unit-ball argument. $\blacksquare$

REFERENCES

- [1] P. M. Frank, "Fault diagnosis in dynamic systems using analytical and knowledge-based redundancy—A survey and some new results," *Automatica*, vol. 26, no. 3, pp. 459–474, 1990.
- [2] J. Gertler, *Fault Detection and Diagnosis in Engineering Systems*. New York: Marcel Dekker, 1998.
- [3] J. Chen and R. J. Patton, *Robust Model-Based Fault Diagnosis for Dynamic Systems*. Boston: Kluwer Academic Publishers, 1999.
- [4] S. X. Ding, *Model-Based Fault Diagnosis Techniques: Design Schemes, Algorithms, and Tools*, 2nd ed. London: Springer, 2013.
- [5] M. Blanke, M. Kinnaert, J. Lunze, and M. Staroswiecki, *Diagnosis and Fault-Tolerant Control*, 3rd ed. Berlin: Springer, 2016.
- [6] R. Dunia and S. J. Qin, "Subspace approach to multidimensional fault identification and reconstruction," *AIChE Journal*, vol. 44, no. 8, pp. 1813–1831, 1998.
- [7] H. H. Yue and S. J. Qin, "Reconstruction-based fault identification using a combined index," *Industrial & Engineering Chemistry Research*, vol. 40, no. 20, pp. 4403–4414, 2001.
- [8] I. Mezić, "Spectral properties of dynamical systems, model reduction and decompositions," *Nonlinear Dynamics*, vol. 41, no. 1–3, pp. 309–325, 2005.

- [9] M. O. Williams, I. G. Kevrekidis, and C. W. Rowley, "A data-driven approximation of the Koopman operator: Extending dynamic mode decomposition," *Journal of Nonlinear Science*, vol. 25, no. 6, pp. 1307–1346, 2015.
- [10] M. Korda and I. Mezić, "Linear predictors for nonlinear dynamical systems: Koopman operator meets model predictive control," *Automatica*, vol. 93, pp. 149–160, 2018.
- [11] B. Lusch, J. N. Kutz, and S. L. Brunton, "Deep learning for universal linear embeddings of nonlinear dynamics," *Nature Communications*, vol. 9, p. 4950, 2018.
- [12] T. Takagi and M. Sugeno, "Fuzzy identification of systems and its applications to modeling and control," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 15, no. 1, pp. 116–132, 1985.
- [13] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, E. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 5998–6008.
- [14] S. X. Ding, P. Zhang, A. Naik, E. L. Ding, and B. Huang, "Subspace method aided data-driven design of fault detection and isolation systems," *Journal of Process Control*, vol. 19, no. 9, pp. 1496–1510, 2009.
- [15] M. Bakhtiaridoust, M. Yadegar, and N. Meskin, "Data-driven fault detection and isolation of nonlinear systems using deep learning for Koopman operator," *ISA Transactions*, vol. 134, pp. 200–211, 2023.
- [16] F. N. Irani, M. Yadegar, and N. Meskin, "Koopman-based deep iISS bilinear parity approach for data-driven fault diagnosis: Experimental demonstration using three-tank system," *Control Engineering Practice*, vol. 142, p. 105744, 2024.
- [17] R. Isermann, "Model-based fault-detection and diagnosis—Status and applications," *Annual Reviews in Control*, vol. 29, no. 1, pp. 71–85, 2005.
- [18] A. Emami-Naeini, M. M. Akhter, and S. M. Rock, "Effect of model uncertainty on failure detection: The threshold selector," *IEEE Transactions on Automatic Control*, vol. 33, no. 12, pp. 1106–1115, 1988.
- [19] P. M. Frank and X. Ding, "Survey of robust residual generation and evaluation methods in observer-based fault detection systems," *Journal of Process Control*, vol. 7, no. 6, pp. 403–424, 1997.
- [20] S. X. Ding, T. Jeinsch, P. M. Frank, and E. L. Ding, "A unified approach to the optimization of fault detection systems," *International Journal of Adaptive Control and Signal Processing*, vol. 14, no. 7, pp. 725–745, 2000.
- [21] A. Johansson, M. Bask, and T. Norlander, "Dynamic threshold generators for robust fault detection in linear systems with parameter uncertainty," *Automatica*, vol. 42, no. 7, pp. 1095–1106, 2006.
- [22] T. Alamo, J. M. Bravo, and E. F. Camacho, "Guaranteed state estimation by zonotopes," *Automatica*, vol. 41, no. 6, pp. 1035–1043, 2005.
- [23] S. Montes de Oca, V. Puig, and J. Blesa, "Robust fault detection based on adaptive threshold generation using interval LPV observers," *International Journal of Adaptive Control and Signal Processing*, vol. 26, no. 3, pp. 258–283, 2012.
- [24] C. Combastel, "Zonotopes and Kalman observers: Gain optimality under distinct uncertainty paradigms and robust convergence," *Automatica*, vol. 55, pp. 265–273, 2015.
- [25] J. Li, Z. Wang, Y. Shen, and Y. Wang, "Zonotopic fault detection observer design for Takagi–Sugeno fuzzy systems," *International Journal of Systems Science*, vol. 49, no. 15, pp. 3216–3230, 2018.
- [26] Y. Ma, Z. Wang, N. Meslem, T. Raïssi, and Y. Shen, "An improved zonotopic approach applied to fault detection for Takagi–Sugeno fuzzy systems," *IEEE Transactions on Fuzzy Systems*, vol. 31, no. 11, pp. 3762–3774, 2023.
- [27] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. New York: Springer, 2005.
- [28] R. Laxhammar and G. Falkman, "Inductive conformal anomaly detection for sequential detection of anomalous sub-trajectories," *Annals of Mathematics and Artificial Intelligence*, vol. 74, no. 1–2, pp. 67–94, 2015.
- [29] F. Cai and X. Koutsoukos, "Real-time out-of-distribution detection in learning-enabled cyber-physical systems," in *Proceedings of the ACM/IEEE 11th International Conference on Cyber-Physical Systems (ICCPs)*, 2020, pp. 174–183.
- [30] R. F. Barber, E. J. Candès, A. Ramdas, and R. J. Tibshirani, "Conformal prediction beyond exchangeability," *The Annals of Statistics*, vol. 51, no. 2, pp. 816–845, 2023.
- [31] R. V. Beard, "Failure accommodation in linear systems through self-reorganization," Ph.D. dissertation, Massachusetts Institute of Technology, 1971.
- [32] M.-A. Massoumnia, "A geometric approach to the synthesis of failure detection filters," *IEEE Transactions on Automatic Control*, vol. 31, no. 9, pp. 839–846, 1986.
- [33] J. J. Gertler and D. Singer, "A new structural framework for parity equation-based failure detection and isolation," *Automatica*, vol. 26, no. 2, pp. 381–388, 1990.
- [34] E. Naderi and K. Khorasani, "A data-driven approach to actuator and sensor fault detection, isolation and estimation in discrete-time linear systems," *Automatica*, vol. 85, pp. 165–178, 2017.
- [35] Y. Fu and W. Yan, "One masked model is all you need for sensor fault detection, isolation and accommodation," in *Proceedings of the International Joint Conference on Neural Networks (IJCNN)*, 2024.
- [36] D. Eriksson, E. Frisk, and M. Krysander, "A method for quantitative fault diagnosability analysis of stochastic linear descriptor models," *Automatica*, vol. 49, no. 6, pp. 1591–1600, 2013.
- [37] R. Sharifi and R. Langari, "Isolability of faults in sensor fault diagnosis," *Mechanical Systems and Signal Processing*, vol. 25, no. 7, pp. 2733–2744, 2011.
- [38] M. Taheri, H. Han, S.-J. Chung, and F. Y. Hadaegh, "Geometric fault identification via mirror descent learning," *arXiv preprint arXiv:2605.17103*, 2026.
- [39] L. L. Scharf and B. Friedlander, "Matched subspace detectors," *IEEE Transactions on Signal Processing*, vol. 42, no. 8, pp. 2146–2157, 1994.
- [40] O. Ledoit and M. Wolf, "A well-conditioned estimator for large-dimensional covariance matrices," *Journal of Multivariate Analysis*, vol. 88, no. 2, pp. 365–411, 2004.
- [41] J. Lei and L. Wasserman, "Distribution-free prediction bands for non-parametric regression," *Journal of the Royal Statistical Society: Series B*, vol. 76, no. 1, pp. 71–96, 2014.